INCH-POUND

MIL-C-24643/43D
22 November 1994
SUPERSEDING
MIL-C-24643/43C
14 March 1994

MILITARY SPECIFICATION SHEET

# CABLE, ELECTRICAL, 600 VOLTS TYPES, LS2SJ, LS3SJ AND LS4SJ (INCLUDING VARIATION LS2SJA, LS3SJA AND LS4SJA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LSSJ (unarmored) and LSSJA (armored).

REQUIREMENTS:

Qualification required.

## Construction (non-watertight)

First - Copper conductor, tin coated, ASTM B 286, sizes 14 and smaller. Uncoated for AWG sizes 12, class B stranding and larger.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Ethylene propylene rubber or cross-linked polyethylene insulation. (see table I for dimensions). Standard identification code applied by method 3.

Fourth - Two, three or four conductors, as required, cabled together with fillers to form a firm, well rounded assembly.

Fifth - Braided shield of tin coated copper. Minimum coverage of 85 percent.

Sixth - Binder tape of polyester, type G in accordance with MIL-I-631, applied helically with overlap.

## Construction variant for type LSSJ

Seventh - Cross-linked polyolefin jacket. (Minimum average wall thickness 0.035 inch.)


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Construction variant for type LSSJA

Seventh - Cross-linked polyolefin jacket. (Minimum, average wall thickness 0.035 inch).
Eighth - Braided metal armor

TABLE I. Details.

| Military part no. M24643/43 1/ | Conductors size | | | | Insulation wall thickness (min avg) (inch) | Lay of conductor maximum (inches) | Type's overall diameter (inches) | | | | Conductor resistance |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Size 2/ | No. | Size 3/ | | | | LSSJ | | LSSJA | | |
| | | | ASTM B 286 | ASTM B 8 | | | min | max | min | max | |
| -01UO | 22 | 2 | 22-19 | | 0.016 | 1-1/2 | 0.261 | 0.275 | 0.311 | 0.325 | 16.54 |
| -02UO | 20 | 2 | 20-19 | | .016 | 1-1/2 | .273 | .290 | .323 | .340 | 10.16 |
| -03UO | 18 | 2 | 18-19 | | .016 | 1-1/2 | .295 | .310 | .345 | .360 | 6.47 |
| -04UO | 16 | 2 | 16-19 | | .016 | 2-1/2 | .309 | .325 | .359 | .375 | 5.02 |
| -05UO | 14 | 2 | 14-19 | | .016 | 2-1/2 | .337 | .350 | .387 | .400 | 3.17 |
| -06UO | 12 | 2 | | 12(class B) | .024 | 3 | .417 | .430 | .467 | .480 | 1.72 |
| -07UO | 11 | 2 | | 10(class B) | .025 | 3 | .447 | .460 | .495 | .510 | 1.085 |
| -08UO | 9 | 2 | | 9(class B) | .031 | 4 | .525 | .545 | .575 | .595 | .86 |
| -09UO | 7 | 2 | | 7(class B) | .032 | 4 | .600 | .615 | .650 | .665 | .54 |
| -10UO | 22 | 3 | 22-19 | | .016 | 1-1/2 | .271 | .285 | .321 | .335 | 16.54 |
| -11UO | 20 | 3 | 20-19 | | .016 | 1-1/2 | .284 | .300 | .334 | .350 | 10.16 |
| * -12UO | 18 | 3 | 18-19 | | .016 | 1-1/2 | .308 | .325 | .358 | .375 | 6.47 |
| -13UO | 16 | 3 | 16-19 | | .016 | 2-1/2 | .323 | .340 | .373 | .390 | 5.02 |
| -14UO | 14 | 3 | 14-19 | | .016 | 2-1/2 | .353 | .370 | .303 | .320 | 3.17 |
| -15UO | 12 | 3 | | 12(class B) | .024 | 3 | .440 | .455 | .490 | .505 | 1.72 |
| -16UO | 9 | 3 | | 9(class B) | .031 | 4 | .594 | .620 | .644 | .670 | .86 |
| * -17UO | 20 | 4 | 20-19 | | .016 | 1-1/2 | .303 | .320 | .353 | .370 | 10.16 |
| -18UO | 16 | 4 | 16-19 | | .016 | 2-1/2 | .346 | .360 | .396 | .410 | 5.02 |
| -19UO | 14 | 4 | 14-19 | | .016 | 3 | .380 | .395 | .430 | .445 | 3.17 |

1/ Unarmored, overall shield. When armored option is required, see MIL-C-24643 for configuration of part number.

2/ Type LSSJ and LSSJA.

3/ Size designation does not correlate to conductor size for all types and sizes.



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EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| Basic electrical: | |
| Conductor resistance - omhs/1000 feet at 25°C, maximum | (see table I) |
| Voltage withstand - volts, root mean square, minimum | |
| Conductor to conductor | 1200 |
| * Conductor to shield | 600 |
| Insulation resistance - Megohms/1000 feet minimum. | |
| Conductor to shield | 500 |
| Jacket flaws | No failure |
| Conductor and shield continuity | No failure |
| Group A: | |
| Visual and dimensional | No failure |
| Group B: | |
| Cross-linked proof test (percent, maximum) | |
| Insulation | 50 |
| Jacket (When tested at 200°C) | 50 |
| Tear - pounds per inch thickness, minimum (ASTM D 470) | 35 |
| Physicals (unaged) | |
| Insulation | |
| Ethylene propylene rubber | |
| Tensile strength - $lb/in^2$, minimum | 700 |
| Elongation - percent, minimum | 250 |
| Cross-linked polyethylene | |
| Tensile strength - $lb/in^2$, minimum | 1800 |
| Elongation - percent, minimum | 250 |
| Jacket (cable) | |
| Tensile strength - $lb/in^2$, minimum | 1300 |
| Elongation - percent, minimum | 160 |
| Group C: | |
| Physicals (aged) | |
| Insulation | |
| Ehylene propylene | |
| Air Oven | |
| Tensile strength - percent of unaged, minimum | 75 |
| Elongation - percent of unaged minimum | 75 |
| Cross-linked polyethylene | |
| Air Oven | |
| Tensile strength - percent of unaged, minimum | 80 |
| Elongation - percent of unaged minimum | 80 |



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| | |
|---|---|
| Jacket (cable) | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 60 |
| Elongation - percent of unaged minimum | 60 |
| Hot oil immersion | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Shrinkage | No failure |
| Heat distortion - percent of unaged, maximum | 39 |
| * Permanence of printing (jacket) - cycles, minimum | 125 |
| Armor (type LSSJA only) - conformance to material, construction and coverage | No failure |
| Shield - conformance to material, construction and coverage | No failure |
| * Surface transfer impedance | |
| * Milliohms per meter, maximum | 700 |
| Emp response time - db, minimum | 60 |

Group D:

| | |
|---|---|
| Flame propagation (cable) | No failure |

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

| | Requirements |
|---|---|
| Aging and compatibility (cable) (125 ± 5°C) | No failure |
| Abrasion resistance (jacket) - scrapes, minimum | 75 |
| Acid gas equivalent - percent, maximum | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Halogen content - percent, maximum | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Immersion (jacket) | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Smoke index, maximum | |
| Jacket | 25 |
| Fillers | 45 |
| Insulation | 45 |
| Toxicity index, maximum | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) - (type A) hardness, minimum | 80 |
| Weathering (jacket) | No failure |



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*UNIT ORDERING LENGTHS:* 1000 feet nominal.

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Custodians:
Army - MI
Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
DLA - IS

Preparing Activity:
Navy - SH

Agent:
DLA-IS

(Project No. 6145-2101-43)



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